

ME5412 Robotics for Healthcare Interfaces for Assistive Technologies

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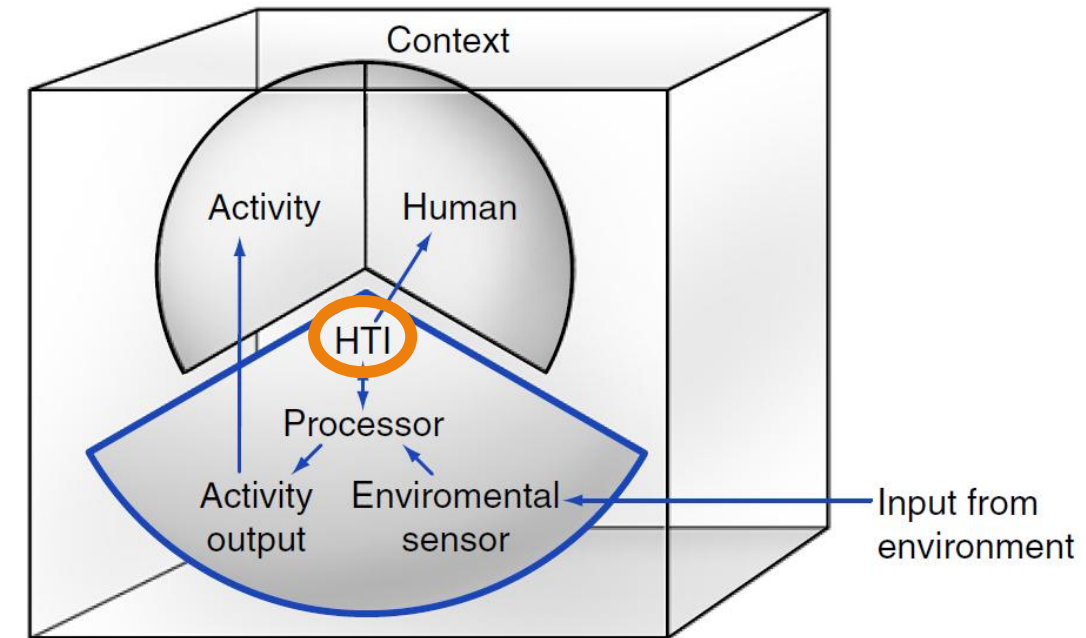
Outline of the Lecture

- Elements of AT interfaces
- Characteristics of AT interfaces
- Adapted interfaces: keyboards, voice recognition, touch screens, eye-gaze systems, pointing devices
- Indirect interfaces
- User input and human skills
- Interfaces for indirect selection: mechanical switches and other switches

Assistive Technology (AT)

Any AT device consists of 4 main components:

1. **Human-AT interface (HTI)**: it can be physical or cognitive, it is bi-directional
2. **Activity output**: it contributes to the functional performance
3. **Processor**: it translates physical (forces) and cognitive (information) inputs into signals used to control the activity output
4. **Environmental sensor**: it detects information on the context and transfers them to the processing unit and to the human, through the human-AT interface



Mainstream vs. Adapted HTIs

- Adapted HTIs are typically used by individuals who have **reduced (fine) motor control** that makes it difficult to use common HTIs such as computer keyboards and pointing devices (e.g., a mouse, or touch screen); light switches; controls on entertainment devices such as TVs; and similar everyday items
- Adapted HTIs are also useful to people who need to use a powered wheelchair or use adapted controls to drive a vehicle

Reduced motor control may depend on:

- Muscle tone (high or low), reduced strength, reduced endurance, reduced range of motion, presence of tremor and the type of tremor
- Neuromotor impairment, (e.g., CP, TBI, ALS) in which there is an impairment of motor control due to abnormalities of tone,
- SCI where there is limited/absent motor function,
- MD and other dystrophies where there is low endurance and limited ROM,
- Musculoskeletal diseases (e.g., RA) that also affects ROM, endurance, and strength

QUADRIPLLEGIC

MOUSE



<https://www.youtube.com/watch?v=HyfgBEZ4F54&t=179s>

Standard Keyboards with Adaptations

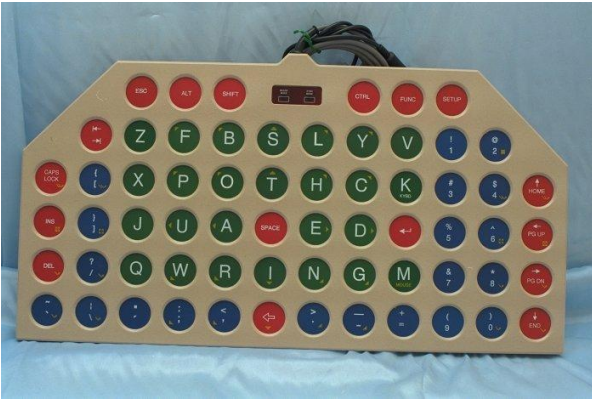
- Keyboard shields, reducing accidental entries



- Ergonomic keyboards



Alternative Keyboards

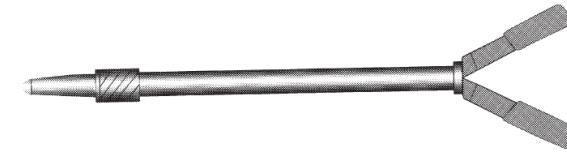


Examples of Alternative Keyboards for Direct Selection

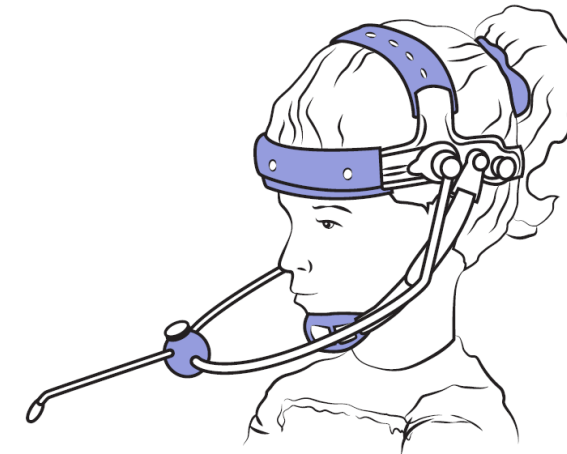
Category	Description	Device Name
Expanded keyboards	Expanded keyboards have enlarged target areas, often programmable so that key size can be customized; useful for individuals with good range and poor resolution; also useful for individuals with limited cognitive or language skills or visual impairment	Big Keys LX, , Early Learning Keyboard, (Ablenet), Jumbo XL II Hi-Visibility Keyboard, HelpiKeys (Inclusive Technologies); Clevy Keyboard (Spectronics)
Contracted keyboards	Miniature, full-function keyboards, typically with membrane overlay; useful for individuals with limited range of motion and good resolution	The Magic Wand Keyboard (In Touch Systems)
Touch screens and touch tablets	Activated by either breaking a very thin light beam or by a capacitive array that detects the electrical charge on the finger; the electrode array used to detect where the finger or pointer is touching is transparent; the touch screen can be placed over the face of a monitor	Touch Windows for various size monitors (Magic Touch);
Special-purpose keyboards	Keyboards on special-purpose devices, such as augmentative communication and environmental control devices; available keys may be much more limited in number or may be specific in function compared with a standard keyboard	

Control Enhancers

- Positioning of user/keyboard
- Pointing devices
- Arm supports



A



B

Control enhancers. A, Mouth stick. B, Head pointer.

Touch Screens

- Optical:
 - Touch interrupts a tiny light beam
- Capacitive:
 - Finger electrical capacitance modifies and is detected in an electric circuit



TouchWindow

Pointing Devices

Required functions:

- Moving a cursor to a target
- Holding the cursor on the target long enough



Alternative Electronic Pointing Interfaces		
Category	Description	Manufacturers
Keypad mouse	Mouse movement is replaced by keys that move the mouse cursor in horizontal, vertical, and diagonal directions. One or more keys perform the functions of the mouse button (click, double-click, drag). The ATEC Computer Switch Interface allows connection of up to five switches to a USB port. There is also a setting for mouse control. The user can use five switches to move the mouse up, down, left, right, and click.	Adaptation, Inc (ATEC Computer Switch Interface)
Trackball	Looks like an inverted mouse; a ball is mounted on a stationary base. Included on the base are one or more buttons that provide the functions of the standard mouse buttons. The base and hand remain stationary, and the fingers move the ball. Requires minimal range of motion and less eye-hand coordination.	Inclusive Technology, Trackman Marble Mouse and Wireless Trackball M570, Logitech, Roller Plus and Roller II Trackball (Traxsys Computer Products)
Continuous input joysticks	Joysticks (continuous input and switched) are used as direct selection interfaces for powered mobility. For computer use, movements are similar to wheelchair control; easy to relate cursor movement (direction, speed, and distance) to joystick movement.	Jouse2 (Compusult Limited); Roller Plus Joystick, Roller II Joystick, and EasiTrax (Traxsys Computer Products); JoyStick-C and JoyStick-C lite (Inclusive Technology); all manufacturers of power wheelchairs have their own joysticks, which are supplied with the wheelchair
Head-controlled mouse	An interface controlled through head movement; the user wears a sensor on the head, which is detected by a unit on the computer. Movement of the head is translated into cursor movement on the screen.	HeadMouse Nano (Origin Instruments Co.); TrackerPro (AbleNet); SmartNAV-4 (Inclusive Technology)
Camera mouse	A camera tracks a body feature (e.g., the tip of the nose), the movement of the feature being used to control an on-screen mouse cursor	CameraMouse (Boston College)
Switch mouse	A scanning strategy, controlled by one or two switches, is used to go over the on-screen targets	One Switch Mouse (Webbie); ScanBuddy (Applied Human Factors)
Light pointers and light sensors	These devices either emit a light beam that can be used to point to objects or as a control interface, or they receive light and provide an output when the light is reflected from an object or the light beam is interrupted.	(Laser light pointers can be used if the precautions shown in Box 8.1 are followed.)

Speech Recognition

- Discrete: require the user to pause between each word
- Continuous: more natural talking, without major pauses
- Speaker dependent: trained on a specific voice, recognition accuracy improves with training
- Speaker independent: no training needed, vocabulary may be smaller

Eye-gaze Systems

Gaze can be detected by:

- IR reflection: IR camera that detected the reflection of an IR beam by the retina
- Detection of pupil position wrt the eye, by a standard camera



Tobii Dynavox Eye Gaze I-13



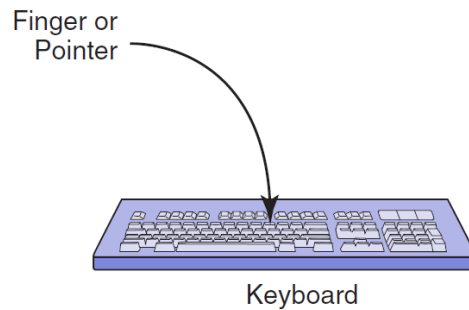
<https://youtu.be/7a39XzTOVac>

Elements of HTIs

- **Control interface:**
 - Hardware by which the user operates or control and AT device
 - Also referred to as **input device**
 - It generates inputs for the AT, in the input domain:
 - Discrete inputs (e.g., keyboard)
 - Continuous inputs (e.g., mouse)
- **Selection set:** group of items available from which choices are made
 - e.g., the keyboard elements, if typing in a computer
 - e.g., controls for moving forward/backward and turning right/left, in a wheelchair
 - Letters, words, symbols, icons, pictures
 - Visual, auditory, tactile (e.g. Braille)
- **Selection method:**
 - Direct selection: the user indicates the choice in the selection set directly
 - Indirect selection: intermediary steps are done for the selection

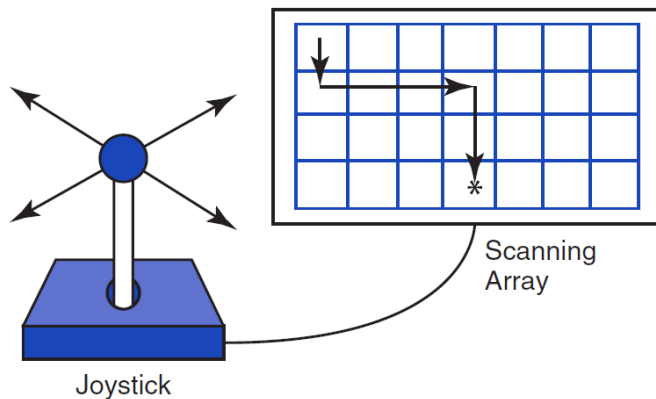
Direct vs. Indirect Selection

Direct Selection



Input	Output
Press S	S

The choice is done directly in the selection set

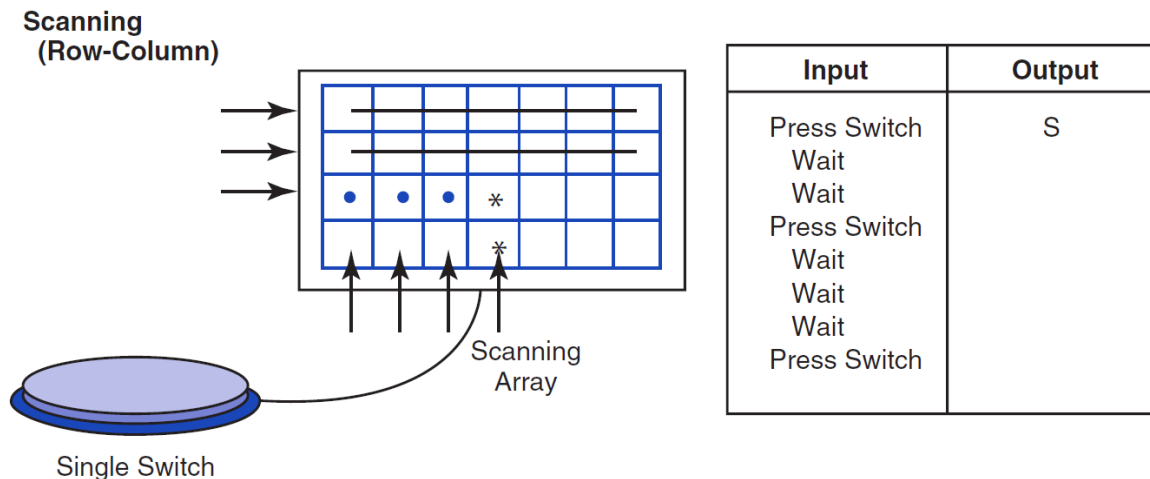


Input	Output
Move Joystick: Down Right Right Right Down	S

The choice is done indirectly through a sequence of steps

Indirect Selection: Scanning

- The selection set is presented on a display
- Each item in the selection set is sequentially lit or indicated by sound or speech
- When the particular element that the user wishes to choose is presented, the user activates a control interface to select that item
- Such control interface is typically a single switch, or an array of two or more switches
- It requires very little motor control

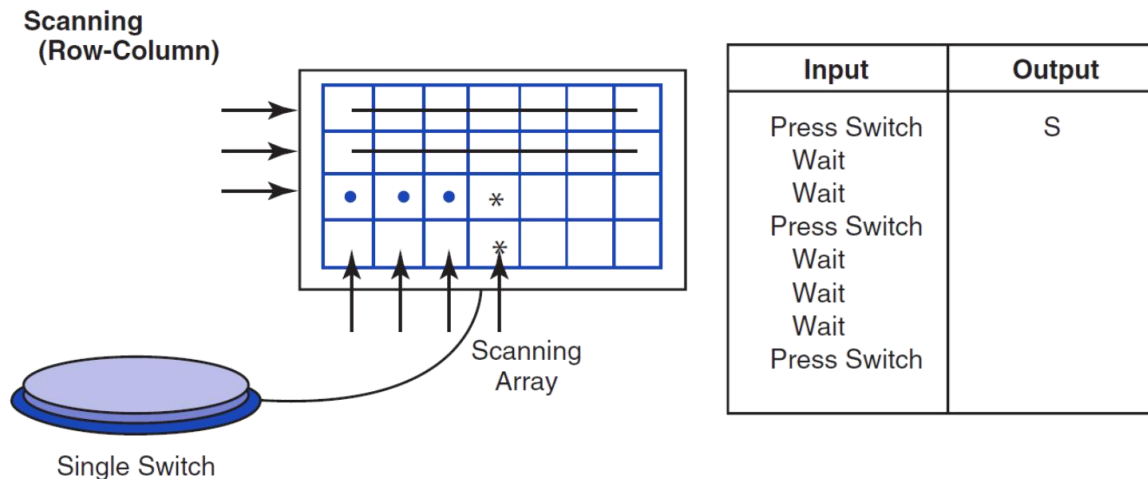


Scanning Techniques

- **Automatic scanning:** items are presented in sequence. The rate of presentation (**scan rate**) can be set and adjusted according to how fast the user can respond. When the desired selection is presented, the user selects the choice by activating the control interface and stopping the scan.
- **Step scanning:** the user activates the control interface once for each item to advance through the choices in the selection set. When the user comes to the desired choice, there are two possibilities for selecting it:
 - an **additional control interface** is used to give a signal to select that choice
 - an **acceptance time** is used. The acceptance time is a slight delay between the time the selection is made and the time it is sent to the device. It allows the user to make a choice by merely waiting for the acceptance time to expire, upon which the selected item is sent to the device.
- **Inverse scanning:** the user initiates the scanning by activating and holding the control interface closed (e.g., keeping a switch pressed). As long as the control interface is held down, the items are scanned. When the desired choice appears, the user releases the control interface to make the selection.

Scanning Formats

- **Linear scanning:** items in the selection are presented horizontally or vertically, one by one
- **Circular/rotary scanning:** items in the selection are presented in a circle, one by one
- **Group-item scanning:** groups of items are scanned initially. The user select the group containing the desired element first. Then, the individual items are scanned and the user selects the item
 - e.g., row-column scanning





<https://www.youtube.com/watch?v=3xPO1yjKGBs>

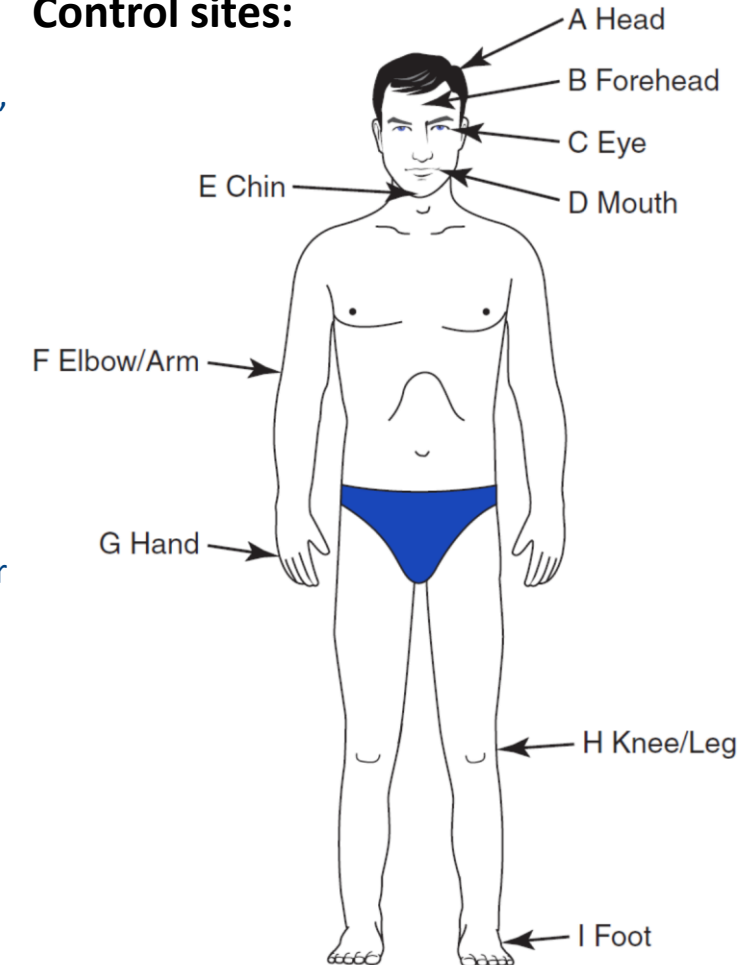
Anatomic Sites for Control of AT

- **hand or finger:** preferred because typically the most dexterous body parts.
 - The fingers can individually flex and extend or, together, perform a grasp and release movement.
 - The thumb can flex and extend, abduct and adduct, and oppose each of the fingers.
- **head:** relatively precise control using head movements such as tilting side to side, horizontal rotation, and tilting up and down. In practice, a combination of them (e.g. rotating left and tilting down)
- If both hand and head control are poor, then adapted HTIs, generally switches, can be used to detect movements of the shoulder, elbow, forearm, hand, or finger.
- **arm, leg:** less desirable for precise tasks because these represent naturally gross movements controlled by large muscle groups.
 - shoulder and scapular movements: elevation/depression, protraction/retraction, internal/external rotation, abduction/adduction, and flexion/extension.
 - elbow movements: flexion and extension
 - forearm movements: pronation (turning the palm down) and supination (turning the palm up)
 - wrist movements: flexion/extension, or from side to side (radial deviation or ulnar deviation).
- **foot:** for fine manipulative tasks, less desirable than the hand or head because visual monitoring can be difficult, and the foot is generally not as finely controlled as the hand. However, some individuals can develop fine control of their feet for typing.
- Control movements: raising and lowering the leg at the hip (e.g., flexion and extension), ankle plantar flexion (toes point down) or dorsiflexion (toes point up), and foot inversion or eversion (rotary movement, similar to pronation and supination).
- **mouth:** for switches based on respiration or phonation
 - respiratory air flow can be detected and used as a control site by sip (inhaling) or puff (exhaling) to access switches
 - phonation may produce sounds (including clicking or whistling) or speech.
- **eye:** gaze movements
- **tongue**

Each control site can perform a variety of movements or actions.

Each of these types of movements can be detected by an appropriate control interface.

Control sites:



Mechanical Switches

Gumball



Mini-Cup



Plate



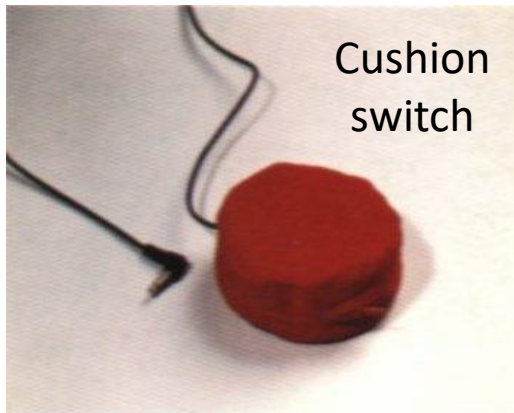
Pneumatic



Adjustable angle switch



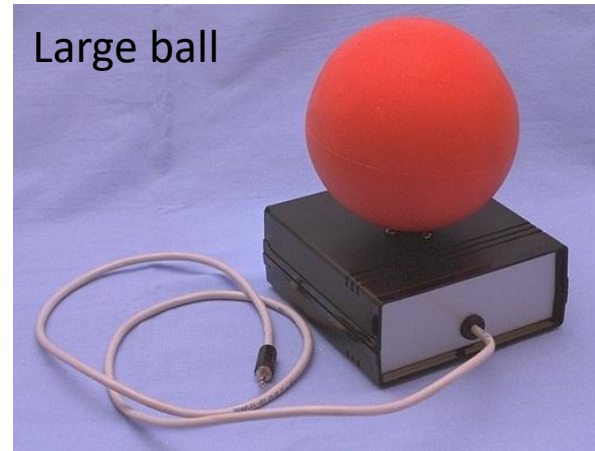
Cushion switch



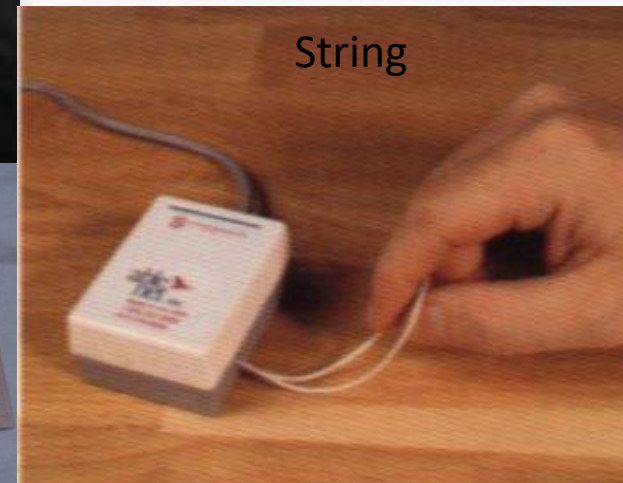
Small ball



Large ball



Mechanical Switches



Other Switches

Microphone



EyeBlink



Using One Switch to Control an iPhone



<https://www.youtube.com/watch?v=bHRIrZxeP8U>

Indirect-selection HTIs

Examples of Single-Switch Control Interfaces		
Category	Description	Switch Name/Manufacturer
Mechanical switches	Activated by the application of a force; generic names of switches include paddle, plate, button, lever, membrane	Pal Pads, Taction Pads, and Flexible Switch (Adaptivation) Corp; Big Red Twist, Jelly Bean Twist, Ribbon Switch, Pillow Switch, Grasp Switch, Wobble Switch, several wireless switches and Buddy Button Switches (AbleNet, Inc.); Lolly, Thumb, Lever (long and short), Leaf (long and short), and Tread Switches (Zygo Industries); Flexible, Wobble, Picture Plate switch (AMD); Egg, Egg w/finger mount, Lip, Button, Treadlite, micro-lite, Ultra Light, Tape Switches (Adaptive Switch Laboratories, Inc.), Cushion Grip, Ultimate (head control), Eyeblink Chin switch, and many other switches with various textured able surfaces, (Enabling Devices)
Electromagnetic switches	Activated by the receipt of electromagnetic energy such as light or radio waves	Fiber Optic Sensor (Adaptive Switch Laboratories, Inc.); SCATIR Minibeamer switch kit (AbleNet, Inc.) Infrared, sound, and touch switch (Words+; now AAC Works)
Electrical control switches	Activated by detection of electrical signals from the surface of the body	BrainFingers (AbleNet, Inc.), EMG Switch (Tinkerton, LLC), Twitch Switch, Switch Enabler kit (Enabling Devices)
Proximity switches	Activated by a movement close to the detector but without actual contact	Candy Corn switch (Ablenet) ASL 204M, 204-3 Pin, 208M, 208-3 Pin Proximity Switch, Honey Bee Switch (Adaptive Switch Laboratories, Inc.); Proximity Switches (AMD)
Pneumatic switches	Activated by detection of respiratory air flow or pressure	Pneumatic Switch (Adaptivation Corp); Sip, Grip and Puff Switch (Toys for Special Children); ASL 308 Pneumatic Switch (Adaptive Switch Laboratories, Inc.); PRC Grip and puff switch (Enabling Devices), Pneumatic Switch (Zygo Industries)
Phonation switches	Activated by sound or speech	Voice Activated and Sound Activated Switches (Enabling Devices); Infrared/Sound and Voice Switches (Zygo Industries), Sound Activated switch (Enabling Devices)

Summary

Questions ?

- Interfaces are a key component of an AT device, since they give access to the device
- As such, they need to be suitable for the user's motor functioning
- Indirect interfaces with scanning mechanisms are a very powerful way to provide access to AT to users with very few controllable motor actions
 - ➔ **they decouple the user's DOF from the device DOF**